M385C
Chapter 1: Sample Spaces and Random Variables
1: Sample Spaces
1.1: Events and sig-alg
Definitions
Sample Space / Outcomes/ Events
sigma-Algebra
Measurable Space
Propositions
(Prop) Closure Properties of sig-alg
Examples
Finest
Coarsest
sig({A})
Countable-Cocountable
Finite-Cofinite Algebra
1.2: Generated sig-alg and Borel sets
Definitions
Generated sig-alg
Borel sig-alg
Propositions
(Prop) \bigcap sig-alg is sig-alg
(Prop) Generator of B(R)
Theorems
(Thm) Existence of non-Borel
2: Random Variables
Definitions
Random Variable
Borel Function
Propositions
(Prop) X^{-1} preserves set ops
(Prop) Generator for Measurability
(Prop) Measurability Characterization for X
(Prop) Compositions of meas. is meas.
(Prop) monotone/cts => Borel
Examples
Every X is RV
Only X=c is RV
\1_{A} is RV <=> A\in \F
X is RV <=> X=c outside C
Corollaries
(postThm) X RV, f Borel => Y=f(X) RV
3: Products and Random Vectors
3.1: Products of sample spaces
Definitions
Product Measurable Space
Propositions
(Prop) \S \circtime \T is smallest sig-alg <=> \pi_S \pi_T meas.
(Prop) f: R—>SxT meas. <=> \pi_S\circ f , \pi_T\circ f meas.
(Prop) Separate meas. =\=> Joint meas.
Theorems
(Thm) B(R^d) = B(R)^{\circtime d}
3.2: Random Vectors, Processes, and Elements
Definitions
Random Vector
Random Element
Random Sequence
Stochastic Process
Propositions
(Prop) X=(X_1,…,X_d) <=> every X_j is RV
(Prop) X_1, …, X_d RVs, f: R^d —> R^m Borel => f(X_1,…,X_d) is R^d-Random Vector
Corollaries
(postThm) Finite Linear Combinations, Products, Maxima, Minima of RVs are RVs
4: Extended Random Variables
Definitions
Extended Real Number
Definition
Topology and Metric
Extended Random Variable
Definition
Arithmatic Convention
liminf / limsup
Propositions
(Prop) Element in B(\bar{R})
(Prop) nonempty A => supA \in \bar{R}, infA \in \bar{R}
(Prop) {X_n} ERV => supX_n / infX_n / liminf X_n / limsup X_n ERVs, {\o \in \O : lim X_n exists in \bar{R}}, X = lim X_n ERV
Example
Example of Element in B(\bar{R})
Cannot use {Y=Z}
Convergence Reveiw on MT
Chapter 2: Probability Spaces
1: Measure Spaces
1.1: Definition and Example
Measure / Measure Space
Set function / Finitely Additive
Probability Measure / Finite Measure / sig-Finite Measure
Null Sets / Atoms
(Prop) S count => \mu(A) = \sum p(x)
Examples
Uniform Probability on finite set
Dirac
Counting
Finite Additive =\=> sig-Additive
1.2: First Properties of Measures
Setup
(Prop) Other Properties of Measrue
FA / M / D / cts up down/ sig-sub
Remark on A_1 finite
1.3: Borel-Cantelli Lemma
Definitions
liminf
limsup
Propositions
(Prop) liminf A_n \subset limsup A_n + De Margan
Theorems
(Thm) Borel-Cantelli Lemma I
(Thm) Borel-Cantelli Lemma II
Counterexample
2: The Lebesgue Measure
Definitions
Lebesgue Measure on R
\pi-system
\lambda-system
sig => pi / lambda
Propositions
(Prop) Lebesgus on intervals
(Prop) \mu_1 \mu_2 on (S,\S), P \pi \sig(P)=\S, \mu_1 = \mu_2 in P => \mu_1 = \mu_2 on \S
(Prop) Lebesgue is TI RI
Theorems
(Thm) Existence of lambda on R
(Thm) Dynkin's \pi-\lambda Thm
***(Thm) Uniqueness of Lebesgue Measure on B(R)
Examples
Counterexample if not \pi-system in unique prop
3: Distributions of Random Variables
Definitions
Probability Space
Pushforward Measure
Distribution
Joint Distribution
Discrete RV
Propositions
(Prop) (S,\S,\mu) —f—> (T,\T), f#\mu is measure on T, if prob then prob
(Prop) X:(\O, \F, P) —> (S,\S) RE, g: S—>T meas. => \mu_{g(x)}
(Prop) Atomic Representation of Discrete Dist
Example: Same Marginal Different Joint
4: Probability Measure on R
Definitions
CDF
Support
Propositions
(Prop) F is 1. nondecreasing, 2. [0,1], 3. rcts, 4. —>0,1
(Prop) \mu((a,b]) = F(b) - F(a), \mu({x}) = F(x) - F(x-)
(Prop) supp(\mu) closed \mu(supp(\mu)^c)=0
Theorems
(Thm) F_{\mu_1} = F_{\mu_2} => \mu_1 = \mu_2
(Thm) Characterization of CDF
F=F_\mu
Riesz-Markov-Kakutani
(Thm) Inverse Method
Corollaries
(postThm) CDF cts <=> no point mass prob
(postThm) Every prob measure on R is a RV distribution
Examples
Uniform Distribution
Exponential Distribution
Normal Distribution
Geometric Distribution
Chapter 3: Integration and Expectation
(S,\S,\mu)
1: The Construction of the Integral
L^0 / L^0_+
L^s / L^s_+ / L^s_+([0,\infty])
Definition
Properties
(Prop) \phi \in L^s => \phi (A_k) = \sum c_k 1_{A_k}
(Prop) \phi \in L^s => \phi (A_k) = \sum c_k 1_{\phi=c_k}
Lebesgue Integral on L^s_+ ([0,\infty])
Definition
Properties
(Prop) \int_S \phi d\mu is Well-defined
(Prop) ML Properties
Remark on infty
Lebesgue Integral on L^0_+ ([0,\infty])
Definition
Properties
(Prop) f \in L^s_+ => f \in L^0_+
(Prop) f \in L^0_+ => 1. \mu(A)>0, f=\infty, \int f d\mu = \infty; 2. \mu(A)=0, f=\infty 1_{A}, \int f d\mu = 0
Lebesgue Integral on L^0([0,\infty])
f^+ / f^- / f / |f|
L^1
2: First Properties of the Integral
Propositions
(Prop) MA on L^0([0,\infty])
(Prop) Approximation L^0_+ ([0,\infty]) by L^s_+([0,\infty])
Proposition
(postThm) MCT => L^s_+ —> L^0_+
Illustration
(Prop) Linearity on L^0_+ ([0,\infty])
Proposition
(postThm) Beppo-Levi
(Prop) L^1(V(0,\infty)) 1. norm finite, 2. L^1 VS 3. Linear 4. T 5. T
Theorems
(Thm) MCT
Theorem
Counterexample f_n = n1_{(0,1/ n]}
(Thm) Fatou's Lemma
(preThm) f=u-v, u,v \in L^0_+ finite => f \in L^1
Theorem
f_n = n1_{(0,1,n]}
(Thm) DCT
Theorem
Comparison of 3 Thms
L^0_+ increase =
L^0_+ ≤
L^0 D =
3: Null Sets
Definitions
Null Set / Null Function
f=g a.e.
Exceptional Set
A=B a.e.
f_n —> f a.e. / X_n —> X a.s.
Propositions
(Prop) = a.e. is equivalence relation
(Prop) A=B a.e. <=> \mu(A\Delta B)=0
(Prop) \mu-\mu^c trivial is sig-alg
(Prop) f,g \in L^0_+([0,\infty]). f=g a.e. => \int f d\mu = \int f d\mu
Proposition
(postThm) f=g a.e. => f\in L^1 <=> g\in L^1 with integrals equal
(Prop) f\in L^0_+([0,\infty]) => \int f d\mu = 0 <=> f=0 a.e.
(Prop) f_n \in L^0, f_n —> f a.e. => f' = limnif f_n \in L^0and f'=f a.e.
Proposition
Remark
Theorems
(Thm) a.e. MCT
(Thm) a.e. Fatou
(Thm) a.e. DCT
(\O, \F, P)
4: Expectation
Definitions
Expectation
Propositions
(Prop) E is 1. Linear, 2. Monotone, 3. Trianglein 4. E[1_A]=P(A))
Theorem
(Thm) BCT
(Thm) E[g(X)] = \int _R g(x) \mu_X (dx) \in [0,\infty]; L^1 <=> L^1
Theorem
Standard Machine: I—L^s+—L^0+—L^0—L^1
Distribution =\= Density
(postThm) \mu_X = \mu_Y => E[g(X)]=E[g(Y)]
(Thm) Change of Variables for General
Theorem
Special on Prob Space
Examples
Discrete Distribution
Absolute cts Distribution
Homework Observation
(Prop) ess sup _{X\in \A} X = sup_{n≥1} X_n a.s.
(Prop) Invariant Measure T* \mu
Chapter 4: Lebsgues Spaces and Inequalities
(S,\S,\mu)
1: Lebesgue Spaces
Keep Difference on L^p and L^p'
Pseudonorms
Pseudonorm
(Prop) f \in L^1' => ||f||_1=0 <=> f=0 a.e.
L^1 = L^1'/~
(Prop) ||[f]||_{L^1} = ||f||_{L^1'} is norm on L^1
L^1'
Convergence in L^1'
DCT Revisit
(Prop) f_n —> f in L^1' => f_n —> g in L^1' <=> f=g a.e.
L^p'
L^p' Space (0,\infty)
(preThm) (a+b)^p ≤ C_p (a^p+b^p)
(Prop) L^p' is VS
Quasinorm
L^\infty'
esssup / essinf
(Prop) C = esssup f => 1. \mu({f>M})=0 2. \mu({f>b})>0, b<M
L^\infty'
Definition
Example
(Prop) L^\infty' is pseudoNLS
Lebesgue Spaces'
Convergence in L^p'
(Prop) L^\infty' Convergence <=> ess. L^\infty Convergence
2: Inequalities on Measure Spaces
Conjugate Exponents
(Thm) Young
(Thm) Hölder
Theorem
(postThm) Cauchy-Schwarz
(Thm) Minkowski
Theorem
(postThm) [1,\infty] => (L^p', ||.||_{L^p'}) pseudoNVS, (L^p, ||.||_{L^p}) NVS
(Thm) Markov
3: L^p is pseudobanach Space
Definitions
Cauchy Sequence
Banach / pseudobanach
(Thm) [1,\infty] => (L^p', ||.||_{L^p'}) pseudobanach, (L^p, ||.||_{L^p}) Banach
(preThm) conv subseq determines Cauchy limit in pseudometric
Theorem
(postThm) [1,\infty], f_n —> f in L^p' => f_{n_k} —> f a.e.
(Prop) (L^2, <.,.>) Hilbert
(\O, \F, P)
4: Inequalities on Probability Spaces
(Thm) Jensen
(preThm) Convex Supporting Line
Theorem
(postThm) A≥G
Propositions
(Prop) Jensen on Interval
(Prop) \phi s.convex + E[|\phi(X)|] <\infty => Jensen Equality <=> X=E[X] a.s.
(Prop) L^p' Embedding
(Thm) lim ||X||_{L^p'} = ||X||_{L^\infty'}, p —> ||X||_p is nondecreasing [1,\infty]
(Thm) Chebyshev
(postThm) Concentration Inequality and Poly Decay
Variance and Properties
Theorem
5: Convergence in Probability
Definitions
a.s. convergence
L^p' convergence
convergence in probability
(Thm) Implications
Examples
L^1' =\=> a.s.
a.s. =\=> L^1'
conv in P =\=> L^1'
Properties
(Prop) Fast Convergence in Prob
Proposition L^0 version
(postThm) ||.||_1 version
(Prop) X_n —> X in L^0 => X_{n_k} —> X a.s.
(Prop) X_n —> X in L^0, X_n —> Y in L^0 => X=Y a.s.
(Thm) X_n —> X in L^0 <=> subsub X_n —> X a.s.
L^0
(Prop) Metrization of conv in prob
(Thm) X_n —> X, Y_n —> Y in L^0, \Phi cts => \Phi(X_n,Y_n) —> \Phi(X,Y) in L^0
Theorem
(postThm) +-*/, max(), min()
(Thm) Egorov
A Glance to understand
Chapter 5: Product of Measure Spaces and Independence
1: Product of Finite Measure Spaces
Sections of Set
Properties on Product Measure
(preThm) \mu_S, \mu_T finite => MMC
Lemma
C = S \circtime T
(Prop) f: SxT —> R in L^0 => f(.,y_0), f(x_0,.) in L^0
(Thm) Product Measure for Finite is unique
Theorem
Finite —> sig-Finite
Theorems
(Thm) Tonelli
Theorem
Standard Extension Machine
(Thm) Fubini
Theorem
(postThm) f \in L^1 <=> \int_S \int_T |f| dxdy < \infty
Comparison
Examples
Without sig-Finite
L^0
(Prop) Lebesgue Measure on R^d Unique, TI, RI
Layer-Cake
(Prop) Layer-cake Formula
(postThm) Moments Tail Formula
(Prop) Polynomial Tail
2: Independence
Definitions
Independent Families
Independent Events
(Prop) A_i indep <=> P(\bigcap A_i) = \Pi_i P(A_i)
Definition
Independent Variable
(Prop) Definition indep X,Y <=> sig(X) indep sig(Y)
Definition
iid
Covariance
Example: Independent => Pairwise Independent
Rademacher RV
Propositions
(Prop) \O \in \A_i => {\A_i} are independent <=> No need subindex
Proposition
\O \notin \A_i => Need subindex
Application of Tonelli-Fubini + Indep
(Prop) X,Y Indep, h: R^2 —> [0,\infty] Borel meas. => E[h(X,Y)] = \int_R^2 … = \int \int… Tonelli-Fubini
(Prop) X,Y Indep => 1. X,Y≥0 => E[XY]=E[X]E[Y] 2. X,Y \in L^1 => E[XY]=E[X]E[Y]
(Prop) X_{ij} indep —Borel meas. f—>Y_i indep
(Prop) X,Y\in L^2 => Cov(X,Y) = E[XY] - E[X]E[Y]
Proposition
Example: Indep => Uncorrelated NOT Revereseed
Theorems
(Thm) {P_i} \pi-sys. indep => {sig(P_i)} indep
(Thm) {Xn} indep <=> \mu_{X1, …, Xn} = \mu_{X1} \circtime …. \circtime \mu_{Xn}
(Thm) {X_n} indep <=> P(X1≤x1, …, Xn≤xn) = \Pi_i P(Xi≤xi)
Theorem
(postThm) Discrete Version
(Thm) Second Borel-Cantelli
3: Selected Reusable Consequences
Propositions
(Prop) (S,\S, \mu) sig-finite, f \in L^0 => \Gamma_f \in B([0,\inf))
(Prop) X,Y \in L^2, (X'-X)(Y'-Y)≥0 => Cov(X,Y)≥0
(Prop) X_n iid => P(X_1 ≤ x)<1 <=> limsup X_n = +\inf a.s.
(Prop) X_n iid => E[|X_1|] < \inf <=> P(|X_n|≥n i.o.)=0
Chapter 6: Derivatives and Densities
1: The Radon-Nikodym Theorem
(Prop) (S,\S,\mu), f: S—>[0,\infty] \in L^0_+ => v(A)=\int_A f d\mu has 3 properties
Definitions
Absolutely Continuous
Equivalent
Mutually Singular
(Prop) Continuity of Finite Measure from <<
Proposition
Finiteness
***Theorems
(Thm) Radon-Nikodym
Radon-Nikodym Derivative
(Thm) Lebesgue Decomposition
Theorem
Example
Calculus on Radon-Nikodym
(Prop) Basic Rules: Linear/ L^0_+ f / Chain Rule / Inverse
(Prop) RN-Derivative of Product Space
Remark
sig-finite necessary
2: Absolutely Continuous RV/V
Definitions
ac RV
ac R.Vector
Density Related
(Prop) Characterization of Density
Proposition
Distribution and Density
(Prop) ac X has no atom
(Prop) <=> Characterization of CDF
(Prop) (X_n) ac => f_{X_k}
Proposition
emark: JJoint => Marginal
(Prop) E(g(X))
3: Sums of Independent RV and *
(Prop) Sums without Independence
Discrete RV
Proposition
Cantor Distribution
(Thm) Z=X+Y, \mu_Z
Proposition
(postThm) CDF of Z
Convolution of Measures
Revisit
(Prop) (\P, *) Abelian Monoid
(Prop) Z=X+Y, f_Z
Proposition
Convolution of Functions
(Prop) (L^1(R), *) Banach Algebra
Chapter 7: Weak Convergence
Definitions
(Thm) Riesz-Markov-Kakutani
Weak-* Convergence in P
Convergence in Distribution
(Prop) 1/ep-Lipschitz on f_{F, \ep}(x)
Proposition
"Mollified Indicator"
(Prop) Weak-* limit is unique
(Thm) Portmanteau
(Prop) Only countable pieces has positive mass
\mu-continuity set
Theorem
Remark
Example
(preThm) Generating \pi-system weak-* conv => weak-* conv
(Thm) CDF Criterion for weak-* conv
Tightness and Compactness
Tight Family
Relative Weak Compactness
(Thm) Prohorov Theorem
Polish Space
Helly's Selection Theorem
(Prop) All {\mu_n_k} has \mu_n_k_l —> \mu in weak-* => \mu_n —> \mu in weak-*
Convergence in Distribution and in Measure
(Prop) X_n —> X in P => X_n —> X in D
(Prop) X_n —> c in D <=> X_n —> c in P
Example
X_n —> X in D \=> X_n —> X in P
Relationship
Homework
(Prop) |X_n| ≤ M, |X| ≤ M a.s. => X_n —> X in D <=> E[X_n^k] —> E[X^k] for all k
Total Variation Convergence
(Prop) TV conv => weak-* conv
(Thm) Scheffé Lemma
Theorem
(postThm) X_n, X ac, f_X_n —> f_X => ||f_X_n - f_X ||_{L^1}—>0
(Thm) Slutsky
Chapter 8: Characteristic Functions
1: First Properties
Definitions
Characteristic Function
MGF
(Prop) Basics of \phi
Proposition
(Thm) Bochner's Theorem
(Prop) Parseval-Type Identity
(Thm) Inversion Theorem
(Prop) Affine Map anf Sums
Theorem
(postThm) \phi_1 = \phi_2 => \mu_1 = \mu_2
(postThm) \phi \in L^1(R) => \mu << \lambda and f(x)= Fourier inversion
Fourier Table for Common
2: Moments and Tails
(Prop) \phi^(k)_X(t) = E[e^{itX}(iX)^k]
Proposition
Converse
(Prop) \mu([-2/d, 2/d]^c) ≤ 1/d \int_{(-d,d)}1-\phi(t) dt
(Prop) \phi(t)-1 = O(|t|^a) => E[|X|^b] < infty for 0<b<a
(preThm) Riemann-Lebesgue
3: The Continuity Theorem
(Thm) Lévy's Continuity Theorem
Theorem
(postThm) \mu_n —> \mu weak <=> \phi_n —> \phi
Continuity at 0
Homework
Multivariable
\phi
(Prop) unique by \phi
(Prop) X_n iid => \phi_{X}= \Pi \phi_X_i
(Prop) Cramér-Wold device
Gaussian Random Vectors
Gaussian RV
(Prop) R^d <—> R, X <=> X \cdot t
(postThm) Affine Transform of Gaussian
(Prop) X_j, X_k Gaussian <=> Cov(X_j, X_k)=0
(Prop) Z ~ N(0,I_d) <=> T(Z) ~ N(m,\Sigma)
(Prop) indep X, X+Y normal => Y normal
Practices
(Prop) \phi (T+t) = \phi (t) => X\in 2\pi/T \Z
(Prop) \phi of Poisson sum
(Prop) Gaussian Smoothing to any Laws
(Prop) P(Y=0)= \pint \phi_Y
(Prop) Cesàro Average Limit = \sum_R \mu({x})^2
(Prop) Gaussian Law is closed under weak-*
(Prop) \mu_n —> \mu in w, |\phi_n| ≤ g \in L^1 => ||f_n-f||_{infty} —> 0
Chapter 9: Limiting Theorems
1. The Weak of Large Number
(Thm) Weak Law of Large Number
(Prop) Cauchy Sample Means
2. The Central Limit Theorem
(preThm) X \in L^2 => t—>0, \phi_X(t) = 1+i\mu t -1/2 m_2 t^2 + o(t^2)
(Thm) Central Limit Theorem
(Thm) Linderberg-Feller
Lindeberg Condition
Theorem
Remark
3. Cycles in Random Permutation
Cycle Decomposition
(Prop) P(L_n=m)=1/n
(Prop) Construction of unif permutation
(Prop) CLT for number of cycles
Homework
(Prop) Poisson Scaling at a=1/2
(Prop) Y_n = \Pi X_k iid >0, X_1 X^-1_1 bdd => Y_n / E[Y_n] —> 0 a.s. <=> X_1 \neq c a.s.
(Prop) Self-normalized Sum CLT
(Prop) Sum of p-Geometric => S_n —> E in D
(Prop) Poisson Sum
(Prop) Convergence of First Moment
(Thm) Stirling Formula
(Prop) Coin biases give different measures
(Thm) Lyapunov CLT
Chapter 10: Conditional Expectation
1. Conditional Expectation
Motivation
Conditional Expectation
(Prop) Conditional Expectation on Finite Partition
Proposition
Example + Tower Property
(Thm) Existence and a.s.-Uniqueness
Theorem
Remark on RN
(Prop) X\in L^2(F) => E[X|\G] = P_{L^2(\G)}[X] Orthogonal Projection))
Proposition
Extension to L^1
2. First Properties
Conditional Expectation Properties [LMIT; JPL; LTI; 3]
(Thm) Linear, Monotone, Identity, Total Expectation
(Thm) Conditional Jensen
(Prop) Pulling out what is known
(postThm) absolute value and L^p nonexpansivity
(Prop) Best L^2 projection
(Thm) Tower Property
Theorem
Information Flow
(Thm) X\in L^1. H indep \sig(G, \sig(X)) => E[X|\sig(G,H)]=E[X| G] a.s.
(Thm) Conditional MCT
(Thm) Conditional Fatou
(Thm) Conditional DCT
(preThm) Doob-Dynkin
Lemma
Conditioning on Y=y
Nonuniqueness of h(y) without a.e.
(Prop) X, Y indep, f: R^2—>R bdd and Borel => g(y)=E[f(X,y)] g(Y)=E[f(X,Y) | \sig(Y)] a.s. Borel
3. Regular Conditional Distributions
Definitions
Conditional Probability
Vector Measure
Measurable Kernel
Regular Conditional Distribution
Standard Borel Space
Conditional F/f
(Prop) A —> P(A|\G) \in L^1 is vector measure
(Thm) Existence of Regular Conditional Distributions
Theorem
Remark
(Thm) Conditional Density from Joint
Theorem
(postThm) Familiar Density Formula
4. Conditioning Gaussian Vectors
Regular Conditional Characteristic
(Prop) Every RVector has a regular conditional \phi
(Prop) RVector => c.\phi has deterministic version <=> \sig(X) indep of \G
(Thm) Conditional law of Gaussian Vector
Setting
Theorem
Remark
(postThm) Binormal Conditional Formula